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MILITARY SPECIFICATION

DIFFERENTIAL PRESSURE TRANSDUCER EQUIPMENT

(ELECTRICAL) (NAVAL SHIPBOARD USE)

1. SCOPE

1.1 Scope. This specification covers the requirements for electrical output differential pressure transducer equipment for Naval ships but does not include the readout or display.

1.2 Classification.

1.2.1 Part classification. The part classification shall be of the following format as specified (see 6.3):

PDT	D	SG	ST	15	WCD	1500	A
Type (see 1.2.1.1)	Sensing technique (see 1.2.1.2)	Transduction principle (see 1.2.1.3)	Application (see 1.2.1.4)	Range (see 1.2.1.5)	Units (see 1.2.1.6)	Pressure rating (see 1.2.1.7)	Current output (see 1.2.1.8)

1.2.1.1 Type. Transducer shall be designated by the three-letter symbol as follows:

PDT - Pressure, differential, transducer.

1.2.1.2 Sensing technique (force summing devices). The sensing technique shall be designated by the corresponding letter symbols and are limited to the following:

D - Diaphragm.
B - Bellows.
T - Straight tube.

1.2.1.3 Transduction principle. The transduction principles shall be designated by the corresponding symbols and are limited to the following:

SG - Strain gage.
VR - Variable reluctance.
DT - Differential transformer.

1.2.1.4 Application. The fluid pressure to be measured shall be indicated as follows:

ST - Steam.
OS - Oil.
SW - Seawater.
FG - Flue gas and ammonia.
OX - Oxygen.
GW - Other gases, fresh water and condensate.

1.2.1.5 Range. The full scale differential pressure range in pounds per square inch differential (psid), or inches of water column differential (WCD) shall be designated by its numerical value (see 3.5.3).

1.2.1.6 Units. The units shall be designated by the corresponding letter symbols and are limited to the following:

Psid - Pounds per square inch, differential.
WCD - Inches of water column, differential.

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1.2.1.7 Pressure rating. The pressure rating shall be designated by its numerical value and is limited to the following as specified (see 6.3):

Pressure rating (psi)
150
600
1500
3000
6000

1.2.1.8 Current output. The current output shall be one of the following (see 3.5.4):

A - 4 to 20 ma. (d.c.)
B - 10 to 50 ma. (d.c.)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements For.
MIL-I-983 - Interior Communication Equipment, Naval Shipboard; Basic Design Requirement For.
MIL-D-1000 - Drawings, Engineering and Associated Lists.
MIL-D-1000/2 - Drawings, Engineering and Associated Lists.
MIL-M-15071 - Manuals, Equipment and Systems.
MIL-P-15137 - Provisioning Technical Documentation for Repair Parts for Electrical and Mechanical Equipment (Naval Shipboard Use).
MIL-S-19500 - Semiconductor Devices; General Specification For.
MIL-T-45208 - Inspection System Requirements.
MIL-C-45662 - Calibration System Requirements.
MIL-P-52211 - Parts and Assemblies, For Generating Plants, Oxygen-Nitrogen, and Related Equipment, Packaging.
MIL-P-55110 - Printed Wiring Boards.

STANDARDS

MILITARY

MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.
MIL-STD-167 - Mechanical Vibrations of Shipboard Equipment.
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-275 - Printed Wiring for Electronic Equipment.
MIL-STD-278 - Fabrication, Welding and Inspection of Machinery, Piping and Pressure Vessels For Ships of the United States Navy.
MIL-STD-701 - Lists of Standard Semiconductor Devices.
MIL-STD-749 - Preparation and Submission of Data for Approval of Nonstandard Electronic Parts.
MIL-STD-758 - Packaging Procedures for Submarine Repair Parts Utilizing Transparent, Flexible, Heat Sealable Film.
MIL-STD-831 - Test Reports, Preparation of.
MS16142 - Boss, Gasket Seal Straight Thread Tube Fitting, Standard Dimensions for.

HANDBOOK

MILITARY

MIL-HDBK-H-1 - Cross-Index of Chemical Equivalent Specifications and Identification Code (Ferrous and Nonferrous Alloys).

DRAWINGS

NAVSHIPS

810-1385846 - Oxygen System, Submarine Replenishment.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Blvd. Chicago, Illinois 60606).

UNITED STATES OF AMERICAN STANDARDS INSTITUTE (USAS)
Y32.16 - Electrical and Electronic Reference Designations.

(Application for copies should be addressed to the United States of America Standards Institute, 10 East 40th Street, New York, New York, 10016).

INSTRUMENT SOCIETY OF AMERICA
Recommended Practice 37.1 - Nomenclature and Specification Terminology for Aerospace Test Transducers with Electrical Ouptput.

(Application for copies should be addressed to the Instrument Society of America, Penn Sheraton Hotel, 530 William Penn Place, Pittsburg, Pennsylvania 15219.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Qualification. The differential pressure transducers furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at time set for opening of bids (see 4.3 and 6.4).

3.2 Definitions. Terminology consistent with Instrument Society of America Recommended Practice 37.1, "Nomenclature and Specification Terminology for Aerospace Test Transducers with Electrical Output" shall apply, except as modified by the definitions listed as follows:

(a) Deadband. The range through which the measurand can be varied without a change in transducer output.
(b) Static error band. The maximum deviation from a straight line drawn through the coordinates of the lower span limit at specified transducer output, and the upper span limit at specified transducer output expressed in percent of transducer range.
(c) Common mode pressure. The static line pressure applied simultaneously to both pressure sides of the transducer.
(d) Overpressure. The maximum expected differential pressure, above or below zero, expressed in percent of differential range of unit.

3.3 Design objectives. The basic design objectives are that the equipment will meet the needs of the Naval service and that the final product will reflect the utmost in simplicity, have maximum reliability consistent with the state of the art, and be easy to install and maintain.

3.4 Description. The equipment specified herein shall be designed to convert a fluid differential pressure to a continuous linear proportional analog electrical direct current (d.c.) signal. The differential pressure transducer shall consist of the following units:

(a) Detector to sense the fluid differential pressure.
(b) Power supply to provide excitation energy to the detector and signal conditioner.
(c) Signal conditioner to convert the detector output to a continuous linear proportional analog electrical signal.

3.4.1 The transducer may be supplied as separate units or may be packaged in any combination.

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3.5 General features. The transducer equipment shall be in accordance with the following paragraphs of MIL-I-983 in addition to the requirements specified herein. Whenever a requirement of MIL-I-983 conflicts with a requirement of this specification, the requirement of this specification shall govern.

Materials, general
Substitution of (equal or superior) materials or parts
Fungus - inert materials
Flammable materials
Arc-resistant materials
Toxic materials
Wood
Metals
Aluminum
Magnesium
Ferrous alloys
Nonferrous material ((except aluminum)
Springs (material)
Other metals
Plastics
Ceramics
Impregnating, embedding and encapsulating compounds
Lubricants and lubrication
Painting
Protection against corrosion
Bolts, machine screws, studs and nuts
Parts - mechanical
Gaskets
Locking devices
Washers
Parts, electrical - general
Use of nonstandard parts
Capacitors
Variable resistors
Transformers and inductors
Minimum sheet metal thicknesses (watertight enclosure)
Through bolting
Cable entrance (stuffing tubes)
Ventilation
Size (for surface ships and submarines)
Threaded devices
Rounded corners and edges
Internal subassembly protection
Drilled and tapped holes
Shock and vibration mountings
Safety (peronnnel hazard)
Shielding and radio frequency noise reduction
Ground potential and grounding
Soldering
Electrical parts mounting
Electrical connectors
Terminal boards and terminals
Wiring
Color coding
Electrical insulation
Dielectric strength and insulation resistance clearances (see 3.11.18 and 3.11.19)
Interchangeability and standardization
Repair parts (see 3.10).
Designation and marking (part designation and serial numbers are required on identification plates)
Workmanship

3.5.1 Materials. Except where specifications are referenced, materials shall be in accordance with commercial specification having material compositions considered suitable for the intended service. Where commercial specifications are referenced equivalent materials shall be in accordance with MIL-HDBK-H1.

3.5.2 Sensing elements. The materials for the force summing devices and wetted parts on both the high and low pressure sides shall be in accordance with table I.

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Table I - Material.

Service	Nickel-copper	Inconel	Copper-nickel	Alloy steel	Corrosion resisting steel	Ni span
Steam	X	X	X	X	X	X
Oil	X	X	X	X	X	X
Seawater	X	X	X	---	---	---
Flue gas and ammonia	---	---	---	X	X	---
Oxygen	X	---	X	---	---	---
Other gases, fresh water, and condensate	X	X	X	---	X	X

3.5.3 Ranges. Transducer differential pressure range shall be as specified (see 6.3). Ranges shall be in accordance with table II.

Table II - Ranges.

Type PDT	
Differential pressure ranges psid	Differential pressure water column ranges inches WCD
0-15	0-10
0-30	0-60
0-60	0-150
0-100	0-300
0-200	
0-400	

3.5.4 Output. Transducer electrical signal output shall be d.c. proportional to transducer differential pressure range. The output shall be one of the following:

A - 4 to 20 milliamperes (ma.) d.c. into an external resistive load of 550 ± 10 percent ohms.
B - 10 to 50 milliamperes (ma.) d.c. into an external resistive load of 600 ± 10 percent ohms.

3.6 Transducer equipment design requirements. Transducer shall be designed for an expected life of 40,000 hours of operation when installed and subjected to the following conditions encountered in Naval service:

(a) High shock (see 4.8.21).
(b) Vibration (see 4.8.20).
(c) Temperature (see 4.8.11) transducer ambient temperature operating range: 5° to 65°C.
(d) Humdity (see 4.8.12).
(e) Supply voltage and frequency variations (see 4.8.7 and 4.8.10).
(f) Inclination (see 4.8.6).
(g) Differential pressure variations in the fluids from over range specified (see 4.8.17).
(h) Common mode pressure at pressure rating (see 4.8.8).

3.6.1 Calibration means. A means shall be provided at the signal conditioner to monitor transducer electrical output corresponding to differential pressure input to permit in place calibration by one man. The signal from this means shall be of an accuracy sufficient to determine the static error band (see 3.11.1) and repeatability (see 3.11.2). The monitoring means may be in the form of test jacks. Monitoring shall be effected without the necessity of electrical disconnection.

3.6.2 Adjustments. Tamper-proof adjustments for zero and span shall be provided for calibration purposes. The number of adjustments shall be kept to a minimum consistent with the operation and maintenance of the equipment and the elimination for the selective matching of parts.

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3.6.3 Enclosure. The differential pressure transducer shall be splashproof as defined in MIL-STD-108.

3.6.4 Enclosure mounting. Differential pressure transducer shall be designed for bulkhead or bracket mounting, as specified (see 6.3).

3.6.5 Size. The differential pressure transducer shall be of necessity, be located on or adjacent to piping, tanks, bulkheads, and so forth, where available space may be at a minimum. In view of these considerations, size should be kept to a reasonable minimum consistent with good engineering practice.

3.6.6 Weight. The weight of the transducer shall be kept to a minimum commensurate with good engineering design.

3.6.7 Differential pressure connections. Transducer differential pressure sensing connections for all services shall be in accordance with MS16142. Only 1/4 inch O.D. tube size shall be used. However, for use on submarine oxygen replenishment systems shown on Drawing 810-1385846, the transducer sensing connection shall be a nickel-copper pipe nipple 1/4 inch IPS with 0.120 inch minimum wall thickness, 6 inches long, welded to the socket in accordance with MIL-STD-278 (see 6.3).

3.6.8 Power supply requirements. The transducer shall be designed to operate with line variations as specified in 4.8.7 and 4.8.10. Nominal steady state power supply requirements shall be 115 volts ± 10 percent, 60 cycles per second ± 5 percent, single phase, unless otherwise specified (see 6.3).

3.6.9 Detector cleaning.

3.6.9.1 Definition. "Clean" shall be defined as being free of all loose scale, rust, grit, filings and other foreign substances and free of mercury, oil, grease or other organic materials. Oil may be present only in detectors for SW, OS and FG applications. The surfaces to be cleaned shall be the pressure connection and all parts contacted by the measurement fluid except for OX where "clean" shall include all surfaces of the detector.

3.6.9.2 Cleaning requirements. All detectors shall be cleaned in accordance with the definitions of 3.6.9.1. In addition the following shall apply:

(a) Detector for OX service shall be clean gas calibrated, cleaned and packaged in accordance with MIL-P-52211.
(b) Detector for GW service shall be clean gas calibrated and pressure connections capped.
(c) Detector for ST service shall be fresh water or clean gas calibrated and pressure connections capped.
(d) Detector for OS, SW, and FG service shall be oil, water or clean gas calibrated and pressure connections capped.

3.6.9.3 Transducer for use with OX and GW service shall have "USE NO OIL FOR CALIBRATION" prominently marked on the detector.

3.6.10 For OX service all joints in the pressure system shall be welded or bolted.

3.6.11 Lubrication. The transducer shall be designed for operation without lubrication of any moving part after assembly. All primary detecting elements shall have the legend "DO NOT LUBRICATE" prominently marked on the detector.

3.6.12 The use of oil for damping, as a fill media, etc., in transducer for OX and GW applications shall required the approval of the Naval Ship Engineering Center (NAVSEC).

3.7 Detail requirements.

3.7.1 Fuses. The transducer shall not be fused.

3.7.2 Semiconductors. Silicon rectifiers, diodes, transistors and other semiconductor devices shall be in accordance with MIL-S-19500 as applicable and shall be selected from MIL-STD-701. Devices other than those listed in MIL-STD-701 shall be used only where specifically approved in writing by NAVSEC on the basis of justification submitted in accordance with requirements of MIL-STD-749.

3.7.3 Electron tubes shall not be used.

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3.7.4 Modular assemblies. The configuration of power supplies, signal conditioners and other circuitry separate from the detector shall meet the modular construction requirements specified in 3.7.4.1, 3.7.4.2 and 3.7.4.3.

3.7.4.1 Module dimensions and packaging, configuration, potting or encapsulation shall be subject to development by the supplier and approved by NAVSEC.

3.7.4.2 The plug in module units may include, but are not limited to, printed circuit boards for conventional components, cable connectors, power supplies and integrated microelectronic circuits (where specifically approved by NAVSEC).

3.7.4.3 Accessibility. Modular units or assemblies shall be fastened in such a manner to allow for quick and easy removal for maintenance, accessibility or replacement.

3.7.5 Printed wiring boards. Printed wiring boards shall be designed and constructed in accordance with MIL-STD-275 and MIL-P-55110.

3.7.6 Resistors. Composition type fixed or variable resistors shall not be used.

3.8 Drawings.

3.8.1 General requirements. Drawings shall conform to MIL-D-1000 and to the detail requirements herein. The number of drawings and the amount of detail shall be held to a minimum consistent with the fulfillment of their function.

3.8.2 Copies of drawings. Unless otherwise specified in the contract or order, two copies of the drawings shall be submitted for initial approval to the NAVSEC. For subsequent procurement the number of copies shall be as specified (see 6.3).

3.8.3 Approval of drawings. Drawings shall be approved by the NAVSEC. Approval shall be required only on the first submittal of an item and whenever changes are made. Where drawing approval has previously been obtained, it shall be necessary only to submit copies of the approved drawing to meet drawing requirements on subsequent contracts. No changes shall be made to approved drawings without prior approval of the NAVSEC.

3.8.4 Master drawings. Master drawings shall be type II, form 2 of MIL-D-1000/2 and shall include the following minimum information, as applicable:

(a) Descriptive data for the equipment.
- (1) Type (see 1.2.1.1).
- (2) Sensing technique (see 1.2.1.2).
- (3) Transduction principle (see 1.2.1.3).
- (4) Application (see 1.2.1.4).
- (5) Range (see 1.2.1.5).
- (6) Units (see 1.2.1.6).
- (7) Pressure rating (see 1.2.1.7).
- # (8) Current output (see 1.2.1.8).
- (9) A statement that the equipment is in accordance with the requirements of this specification and to referenced specifications.
- (10) A list of referenced drawings.
- (11) Description of the nature and purpose of any adjustments.
- (12) Test procedures approval data, including letter, report numbers and dates.
- (13) Shock classification.
- (14) Enclosure classification.
- (15) Ambient temperature.
- (16) Weight of equipment (net).
- (17) Any special features.
- (18) A tabulation of approved transducer equipment covered by this drawing listed by part classification (see 1.2.1).

(b) List of material, presenting following data in tabular form.
- (1) Piece number (for flags of assembly drawing).
- (2) Name of part (for example, transformer, diode, and other similar parts).
- (3) Schematic diagram designation for electrical parts (for example, switch S-1, and other similar designations).
- (4) Number of each part required per assembly.
- (5) Military or Federal Specification number or NAVSHIPS drawing number, if any.
- (6) Type, class, grade, size, military designation or other classification of any referenced specification.
- (7) Part number or identification assigned by manufacturer.
- (8) Name of manufacturer of part.

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(9) Part number or identification assigned by part manufacturer.
(10) Federal stock number, if known.
(11) Rating of part from part manufacturer's catalog (for example, resistor 10 ohm 25 watt; diode - 200 PRV, 750 milliamps and similar ratings.
(12) Remarks column.

(c) Miscellaneous part and component requirements and application data. Information on parts and components shall include the data specified below. The data required shall be presented in tabular form wherever practical. Each part shall be identified by its reference designation.

(1) Transformer, reactor and inductor data:
a. Core data:
1. Lamination form - EI, toroid, and so forth.
2. Lamination material and gage.
3. Stack and interleave dimensions.
4. Construction - open, tape wound cut core, and other construction.
5. Core box, if any.
6. Ground insulation.
b. Winding data.
1. Insulation class - class B, H, and so forth.
2. Winding identification - same terminal designation as given in schematic and connection diagrams.
3. Wire size and insulation type, for example AWG 10 type B2 and so forth.
4. Number of turns.
5. Nominal voltage between terminals.
6. D.c. resistance at 25°C.
7. Material of insulation used between winding layers and between windings.
c. Varnish impregnation
1. Varnish type and specification.
2. Varnish manufacturer and material identification.
3. Impregnation procedure - preheating temperature and time, number of dips and bakes, temperature and time of bakes, and so forth.
d. Volt-ampere ratings for potential transformers and current ratios for current transformers.

(2) Semiconductor data including rating and complete ID for ordering from original supplier (JEDEC number is adequate), and the following:
a. Rectifiers using silicone diodes:
1. Circuit symbols and piece number.
2. Maximum reverse voltage peak and rms (complete rectifier circuit).
3. Maximum load current d.c. amperes (complete circuit).
4. Maximum diode case temperature at design ambient - use for diodes rated case temperature.
5. Maximum ambient temperature directly adjacent to diode - use for diodes rated against ambient temperature - no cooling plates.
6. Cooling plate size, thickness and material where so mounted.
7. Circuit (for example, single phase bridge, and so forth).
8. Total diodes in circuit.
9. Diodes in series per arm.
10. Diodes in parallel per arm.
11. Description of insulation hardware if insulated from a cooling plate.
b. Voltage regulator diodes:
1. Circuit symbol(s) and piece number.
2. Range of actual voltage levels which circuit will accept (with or without circuit adjustment from regulator diodes) and still give required equipment performance_____volts to _____volts.
3. Regulation (including temperature error) permitted in an installed regulator diode under all conditions of operation (assume equipment is initially adjusted for optimum performance) +_____millivolts.
4. Maximum level of steady-state average reverse current through regulator diode (steady-state shall be considered any time in excess of 30 seconds). _____ma.
5. Maximum transient reversed current under any operating condition _____ma for _____sec. duration and _____frequency of occurence.
6. Maximum ambient temperature at diode (no cooling plate).
7. Maximum case temperature of diode (complete for diode rated against case temperature.

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8. Cooling plate size, thickness and material where used.
9. Total diodes in circuit.
10. *Diodes in series per circuit element (complete where more than* one circuit element is used in the voltage regulator such as in a bridge type regulator circuit).
11. Description of insulation hardware if insulated from a cooling plate.

c. Voltage reference diodes (see regulator diodes also).
1. Circuit symbol(s) and piece number.
2. Range of reference voltage levels which circuit will accept (with or without circuit adjustment) from the voltage reference diode(s): _____volts to _____volts.
3. Error permitted in an installed voltage reference diode under all conditions of operation: ± _____millivolts.
4. Circuit design current through voltage reference.
a. Nominal or average value: _____ma.
b. Regulation about nominal value: ± _____ma.
5. Maximum ambient temperature (include internal heat rise above outside design ambient): _____a.
6. Number of voltage reference elements in series: (Operation in parallel is not permitted).

d. Transistors:
1. Circuit symbol(s) and piece number.
2. Maximum collector to emitter voltage ___volts.
3. Maximum collector to base voltage ___volts.
4. Maximum collector current _____ma.
5. Maximum base current _____ma.
6. Maximum collector power dissipation _____mw.
7. Maximum ambient temperature (include internal heat rise at transistor at 65°C., design ambient _____°C).
8. Maximum case temperature of transistor at 65°C. equipment design ambient and any condition of required equipment operation _____°C. (Where rating is based on case temperature.)
9. Cooling plate size, thickness and material where so mounted.
10. Description of insulating hardware if insulated from a cooling plate.

e. Silicon controlled rectifiers:
1. Circuit symbol(s) and piece number.
2. Nominal working peak forward and reverse voltages.
3. Repetitive peak forward and reverse voltages (maximum).
4. Non-repetitive peak forward and reverse voltages (maximum).
5. False firing, or self-firing, restrictions.
6. Repetitive peak forward current (nominal and maximum values).
7. Surge current (peak for non-repetitive value occurring within the conduction portion of a single cycle but after turn-on has occurred, effective amplitude and cycles duration for multi-cycle surges, or amplitude and duration for sub-cycle pulse duration).
8. Peak current and maximum rate of current rise during turn-on.
9. Minimum time from the instant of voltage reversal to the instant when voltage is reapplied in the forward direction.
10. Peak forward gate current (minimum and maximum, per device).
11. Peak forward gate voltage (minimum and maximum, per device).
12. Peak reverse gate voltage (maximum, per device).
13. Maximum gate power dissipation (average and peak, per device).
14. Maximum device case temperature at design ambient.
15. *Cooling plate size, thickness and material, where so mounted.*
16. Circuit.
17. Total devices in circuit.
18. Devices in series per arm.
19. Devices in parallel per arm.
20. Insulation hardware if insulated from a cooling plate.

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f. Selenium surge suppressors:
1. Identify the suppressor manufacturer's technical data bulletin applicable to each product certified for the application. For each application indicate which product is installed in the equipment and which is the alternate product that may be used for replacement purposes, when desired.
2. Provide the following product rating data:
a. Stack rms voltage (steady-state).
b. Stack clamping voltages (new and aged).
c. Peak discharge current (corresponding to clamping voltage).
d. Polarization (polarized or non-polarized.)
e. Maximum energy (watt-seconds) at rated clamping voltage (aged) for single pulse (non-repetitive) current.
f. Maximum pulse duration (micro-seconds) for repetitive pulses at supply frequency (specify cycles per second) and amplitude equal to clamping voltage (aged).
g. Maximum ambient temperature at which rated values in 2a through 2e apply.
h. Maximum operating temperature.

g. Integrated circuits:
1. Manufacturer and manufacturer's type number.
2. Input/output relationship.
3. Maximum allowed terminal voltages and currents.
4. Maximum applied terminal voltages and currents.
5. Maximum allowed and applied power dissipation.
6. Maximum ambient temperature.
7. Maximum case temperature at 65°C, equipment design ambient, at worst case operating conditions.

(3) Resistor data including the type, resistance and wattage ratings, operating temperature and standard Military type number (if applicable), actual power dissipated at design operating temperature.

(4) Capacitor data including the type, capacitance, voltage rating, maximum applied d.c. and a.c. voltage and frequency values (include transient), and standard Military type number (if applicable).

(5) Switch data including the type of switch, contact arrangement, number of poles, number of positions, and voltage and current ratings of contacts.

(d) Electrical wiring and schematic diagrams.

(1) Schematic - A single electrical schematic diagram for each unit to represent clearly the operation of the equipment. The following features shall be incorporated in the diagram:

a. In preparation of the schematic, emphasis shall be placed on simplicity and ease of understanding of circuit operation. Physical placement of components and connecting wiring may be ignored in the interest of simplicity and clarity of this diagram.

b. A thin, broken line shall be used to represent the boundaries of each unit of subassembly. All terminals to which external connections are made shall be shown within these boundaries, with the numbers, markings, type of signal, power and ground, as appropriate.

c. Each part, such as resistors, capacitors, relays shall be given a unique reference designation consisting of a letter denoting the type of part (as required by USAS Y32.16) and a number assigned consecutively. The numbers shall be assigned in a logical sequence of the electrical current or signal flow through the circuit.

d. In addition to the reference designation, the following information shall be noted adjacent to the part. Where numerical values are given, a code shall be noted to designate the units used:
1. Resistors - Resistance, power rating, and tolerance. If variable, an arrow to indicate clockwise rotation of the control shaft.
2. Capacitors - Capacitance, voltage rating, and tolerance.
3. Reactors - Inductance and voltage rating.
4. Semi-conductors - type number.
5. Integrated circuits - Type number, operational symbol, diagram of input/output relationship, terminal numbering corresponding to a representative schematic.
6. Controls, adjustments and test points - the assigned name.

e. All supply voltages, phases and frequencies and transformer terminal voltage shall be indicated and labeled as to purpose.

f. Winding polarities shall be indicated for transformers and saturable reactors. The same system shall be used throughout. The method of marking polarity shall be shown and explained in a note.

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g. All schematic diagrams, size D and larger, shall be zoned. Zone widths shall be between 6 and 8 inches. Smaller size drawings may be zoned for clarity.

h. Description of operation - A brief description of operation of the circuit elements to enable approval authorities and operators to understand circuitry.

(2) Wiring diagram - A wiring diagram shall be drawn for each unit showing components and wiring in correct relation to physical positions. Wiring between parts and interconnection wiring shall be marked and color coded. Markings and part designations shall be consistent with other diagrams provided. Hook-up wire size and type shall be specified. Notes labeling terminals shall be provided for identification.

(e) Constructional details.

(1) Two representative assembly views, as required, to show clearly the details of the design, construction and assembly of the unit and to identify each part and its location. Identification of parts shall correspond to the list of materials.

(2) Overall, drilling and mounting dimensions.

(3) Sectional views or notes as necessary to show or explain special mounting provisions or installation procedures (for example, mounting and module enclosure requirements for separately mounted modules).

(4) Details such as cable entrance provisions, gaskets, special mounting requirements and other details as applicable.

(5) Outline drawing of parts or components furnished separately (not mounted in an enclosure) to show critical dimensions including mounting dimensions.

(f) Installation drawings. Data to be shown on these drawings shall be as follows:

(1) Dimensional outline of all major assemblies (detector, module, and so forth), showing overall and principle dimensions in sufficient detail to establish the limits of space in all directions required for installation, operation and servicing, exclusive of space required for personnel. Include the amount of clearances required for withdrawal of parts and subassemblies, clearances for cable and piping access, and clearances for ventilation. Also indicate the location of operating controls and test points.

(2) A table of reference drawings to include drawing number of each major assembly, weight of each enclosure and separately furnished assembly and heat dissipation of major assemblies.

(3) A block diagram of all major assemblies showing interconnection between these assemblies. Terminal identification, as well as maximum current rating for each cable should be shown.

(4) Location, type and dimensions of cable entrance plates, connectors, differential pressure connections with optional entrances indicated.

(5) Any special instructions for preservation, painting, installation or assembly as necessary.

(6) Heat dissipation.

(g) Drawing format. All of the above information may be presented on one sheet. However, separate sheets may be used for (c), (d) and (f). All other information, as required above, shall be presented on one sheet.

3.9 Manuals. Manuals shall be supplied in the quantity specified (see 6.3) and shall be in accordance with type I of MIL-M-15071 with content and arrangement as follows:

(a) Front matter.
(b) General information.
(c) Installation:
(1) This section is to include complete calibration and alignment procedures.
(d) Operation.
(e) Maintenance, repair and troubleshooting.
(f) Parts identification.
(g) Drawings - Reduced size master drawing inserts.
(h) Appendix - Inserts on special components.

One initial copy of the manual shall be supplied to NAVSEC for approval.

3.10 Onboard repair parts. The onboard repair parts shall be supplied in quantity as specified (see 6.3) and shall be in accordance with MIL-P-15137.

3.11 Transducer performance. Performance tolerances are specified in percent of transducer span unless stated otherwise.

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3.11.1 Static error band. The transducer static error band shall not exceed plus or minus 1 percent (see 4.8.1).

3.11.2 Repeatability. Repeatability of transducer output shall be within 0.5 percent (see 4.8.1).

3.11.3 Threshold and deadband. The maximum least detectable increment plus deadband shall not exceed 0.20 percent (see 4.8.3).

3.11.4 Ripple. Transducer output ripple shall not exceed 0.15 percent of full scale output (see 4.8.4).

3.11.5 Warm-up time. The transducer output shall attain a value within plus or minus 1 percent of the steady-state output with no overshoot in excess of 1 percent. Output shall reach this band in 30 minutes or less and shall remain in this band (see 4.8.5).

3.11.6 Inclination. Maximum deviation of transducer output resulting from inclination shall not exceed 3.0 percent (see 4.8.6).

3.11.7 Supply voltage and frequency (steady-state). Maximum difference between outputs at any voltage and frequency condition and the normal (115 volts, 60 cycle) at the same input, pressure and ambient temperature shall not exceed 1.0 percent (see 4.8.7).

3.11.8 Temperature error. Temperature error based on the test temperature extremes of the detector shall not exceed 0.02 percent per °C. (see 4.8.24).

3.11.9 Common mode pressure. During the common mode pressure test, transducer performance shall be within the static error band specified in 3.11.1 plus

$$1/10 \left(\frac{\text{System operating pressure}}{\text{D/P range}}\right)^{1/3}$$

(see 4.8.8).

3.11.10 Frequency response. Frequency response of the transducer as determined by 4.8.9 shall have a magnitude ratio within plus or minus 0.2 decibels (db) to a frequency of 0.5 c.p.s. The transducer output shall exhibit no magnitude ratio greater than plus 1 db at the damped natural frequency if it occurs within the range of test frequencies.

3.11.11 Supply voltage and frequency (transient) (see 4.8.10).

3.11.11.1 Voltage. During the voltage transient test the transducer output shall remain within plus or minus 5 percent of the steady state output.

3.11.11.2 Frequency. During the frequency transient test the transducer output shall remain within plus or minus 0.5 percent of the steady state output.

3.11.12 Temperature. After the temperature test, transducer performance shall be within the static error band specified in 3.11.1 (see 4.8.11).

3.11.13 Humidity (see 4.8.12).

3.11.13.1 Temperature cycle. Maximum deviation of second cycle reference measurement from the post conditioning calibration shall not exceed 1.5 percent. After temperature cycling, performance shall conform to 3.11.1.

3.11.13.2 Humidity cycle. Maximum deviation of any 30 minute interval output value from the first measurement shall not exceed 1 percent. Equipment shall show no visual evidence of damage (see 4.8.12).

3.11.14 Overpressure. Calibration conducted after overpressure test shall have no values in excess of 1 percent deviation from the pre-overpressure test calibration. After calibration the transducer shall be capable of being adjusted to accuracy requirements of 3.11.1 (see 4.8.13).

3.11.15 Line pressure. After the line pressure test, transducer performance shall be within the static error band specified in 3.11.1 (see 4.8.14).

3.11.16 Salt spray (detector only). No appreciable corrosion or other damage shall be evident after the salt spray test specified in 4.8.16.

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3.11.17 Differential pressure whip. Calibration conducted after completion of differential pressure whip test shall have no values in excess of 1 percent deviation from pretest reference measurement (see 4.8.17).

3.11.18 Dielectric strength. Transducer equipment shall be capable of withstanding dielectric test voltages as specified in 4.8.18. When subjected to this test, the equipment shall show no evidence of punctured insulation or arcing.

3.11.19 Insulation resistance. The insulation resistance of the transducer shall be not less than 10 megohms (see 4.8.19).

3.11.20 Vibration. Monitored transducer output during all phases of vibration test shall show no variation from steady state output in excess of 2.0 percent. There shall be no visible evidence of damage to the transducer as a result of the vibration test (see 4.8.20).

3.11.21 Shock. After the shock test but prior to any adjustment the transducer output shall show no deviation greater than 3 percent. A post test calibration, using adjustments provided, shall conform to performance requirements of 3.11.1. There shall be no visual evidence of damage to the transducer as a result of the shock test (see 4.8.21).

3.11.22 Burst pressure. The transducer shall be capable of withstanding the burst pressure specified in 4.8.22 without showing any evidence of leakage.

3.11.23 Output. The output shall conform to 3.5.4 and the transducer performance shall be within the static error band specified in 3.11.1.

3.11.24 Enclosure. There shall be no evidence of water leakage into the equipment enclosure (see 4.8.15).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection system. The inspection system which the supplier is required to maintain, as provided in the inspection clause of the contract or order shall be in accordance with MIL-I-45208.

4.2 Classification of inspection. Examination and tests shall consist of the following:

(a) Qualification tests (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Periodic tests (see 4.5).

4.3 Qualification tests[1/]. Qualification tests shall be conducted at a laboratory satisfactory to the NAVSEC. Qualification tests shall consist of the examination and tests specified in table III.

Table III - Qualification, examination and tests

Examination and tests	Requirement paragraph	Examination and test paragraph
General examination	3.5	4.6
Output	3.11.23	4.8.23
Warm-up time	3.11.5	4.8.5
Enclosure	3.11.24	4.8.15
Salt spray (detector only)	3.11.16	4.8.16
Static error band and repeatability	3.11.1 and 3.11.2	4.8.1
Threshold and deadband	3.11.3	4.8.3
Ripple	3.11.4	4.8.4

[1/] Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3 and 6.3.1).

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Table III - Qualification, examination and tests (cont'd.)

Examination and tests	Requirement paragraph	Examination and test paragraph
Inclination	3.11.6	4.8.6
Supply voltage and frequency (steady state)	3.11.7	4.8.7
Common mode pressure	3.11.9	4.8.8
Frequency response	3.11.10	4.8.9
Supply voltage and frequency (transient)	3.11.11	4.8.10
Temperature	3.11.12	4.8.11
Temperature error	3.11.8	4.8.24
Humidity	3.11.13	4.8.12
Overpressure	3.11.14	4.8.13
Line pressure	3.11.15	4.8.14
Differential pressure whip	3.11.17	4.8.17
Dielectric strength	3.11.18	4.8.18
Insulation resistance	3.11.19	4.8.19
Vibration	3.11.20	4.8.20
Shock	3.11.21	4.8.21
Burst pressure	3.11.22	4.8.22

4.3.1 Samples for qualification. The number of samples to be submitted for the qualification tests specified in table III shall depend on the transducer design. If each range is covered by a separate and distinct design, a sample for each range will require testing. Only one sample of a basically similar design series which may cover many ranges and types need be submitted for testing if mechanical and electrical similarity is deemed sufficient by NAVSEC. The range which has successfully passed the qualification tests for this design will qualify all other ranges and types of the same design.

4.4 Quality conformance inspection. Each pressure transducer in each lot offered for delivery shall be subjected to the inspection listed on table IV and shall be conducted in the order listed. Failure of any pressure transducer to meet the requirements of this specification shall be cause for rejection.

Table IV - Quality conformance inspection.

Inspection	Requirement paragraph	Examination and test paragraph
General examination	3.5	4.6
Output	3.11.23	4.8.23
Static error band and repeatability	3.11.1 and 3.11.2	4.8.1
Commom mode pressure	3.11.9	4.8.8
Overpressure	3.11.14	4.8.13
Line pressure	3.11.15	4.8.14
Dielectric strength	3.11.18	4.8.18
Insulation resistance	3.11.19	4.8.19

4.4.1 Quality conformance inspection reports. An inspection report shall be submitted with each transducer to attest to the performance of the inspection specified in 4.4. Report shall include a complete identification of the transducer with a check notation of "passed" or "not passed" for each examination and test, the requirement paragraph and applicable test paragraph. A transducer calibration record shall be included to determine conformance with the requirements of 3.11.1 and 3.11.2.

4.5 Periodic tests. Three transducers shall be selected at random out of every 100 transducers which have satisfactorily passed the tests in 4.4, from each lot offered for delivery and shall be subjected to the tests listed in table V. If any transducer fails in any test, no transducers shall be accepted for quality conformance inspection until the manufacturer has determined the cause of the defect and has taken the necessary action to correct or eliminate the defects from all units on hand. The failed test and any other periodic tests required by the Government shall be repeated to demonstrate that the corrective action will enable the transducer to conform to the requirements of this specification. Samples which have satisfactorily passed the examination and tests listed in table V may not be applied as part of the quantity specified (see 6.3).

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Table V - Periodic tests

Test	Requirement paragraph	Test paragraph
Inclination	3.11.6	4.8.6
Supply voltage and frequency (steady state)	3.11.7	4.8.7
Temperature error	3.11.8	4.8.24
Temperature	3.11.12	4.8.11
Dielectric strength	3.11.18	4.8.18
Insulation resistance	3.11.19	4.8.19
Burst pressure	3.11.22	4.8.22

4.5.1 Periodic test reports. A complete report of all periodic tests conducted as specified in 4.5 shall be submitted to the NAVSEC for approval. Format for the reports shall conform to MIL-STD-831.

4.6 General examination. Transducer shall be given a thorough esamination to determine conformance to the requirements of this specification with respect to material, finish, workmanship, construction, assembly, dimensions, weight, marking of identification and information plates. Examination shall be limited to the examinations that may be performed without disassembling the units in such a manner that its performance, durability or appearance would be affected. Examination shall also include a check of all operation controls and adjustments, as applicable.

4.7 Test conditions and test procedures.

4.7.1 Test conditions. Except where the following factors are the variables, the tests specified in 4.8 shall be conducted with the equipment operating under the following conditions:

(a) Ambient temperature shall be 25° ± 2°C.
(b) Relative humidity shall be ambient.
(c) Supply voltage shall be 115 volts (nominal).
(d) Supply frequency shall be 60 c.p.s. (nominal).
(e) Controls shall be in the neutral or normal position.
(f) Specified external load (see 3.5.4).

4.7.2 Calibration system requirements. The supplier shall provide and maintain a calibration system for test equipment. The calibration system shall be in accordance with MIL-C-45662 and shall be acceptable to the Government.

4.8 Tests. The transducer to be tested and all associated test equipment shall be energized for a long enough period of time to ensure complete warm-up. The period of time shall be at least 2 hours. After the warm-up time test, the transducer may be allowed a 1/2 hour warm-up period.

4.8.1 Static error band and repeatability. The transducer shall first be flexed over its full pressure range by slowly increasing and decreasing the applied pressure for 6 continuous cycles. The calibration measurement shall be made at a minimum of 5 equally spaced intervals over the entire range (both upscale and downscale). Precaution shall be taken to avoid overshoot. This calibration procedure shall be applied 3 successive times to determine repeatability. Static error band of all calibration shall meet the requirements of 3.11.1. Repeatability shall meet the requirements of 3.11.2.

4.8.2 Reference measurement. The following reference measurement shall be conducted when specified in the individual test:

(a) A one trial calibration with at least 5 equally spaced intervals over the entire transducer range both upscale and downscale.

4.8.3 Determination of threshold and deadband. Threshold and deadband shall be determined with 80 ± 5 percent of the rated differential pressure applied to the transducer. The differential pressure interval beyond this point required to produce a detectable change in output and the differential pressure interval in the opposite direction to return the output to the original value shall be determined in each direction. Threshold is the first differential pressure interval. The second differential pressure interval is the sum of threshold and deadband. Performance shall conform to the requirements of 3.11.3.

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4.8.4 Ripple. Ripple shall be determined at an input differential pressure of 80 ± 5 percent of transducer span. Performance shall conform to the requirements of 3.11.4.

4.8.5 Determination of warm-up time. Test shall be conducted to determine the elapsed time between the application of line power to the transducer and the point at which the transducer output reaches the conditions specified in 3.11.5.

4.8.5.1 The transducer shall be placed in an ambient temperature of 25° ± 2°C. for not less than 2 hours de-energized. Recording equipment and other auxiliary equipment shall be energized to assure complete warm-up. An input differential pressure of 80 ± 5 percent of the transducer span is to be applied to the transducer and maintained constant during this test. Performance shall conform to 3.11.5.

4.8.6 Inclination. The equipment shall be inclined for a period of at least 1 minute in each of the following positions:

(a) 45 degrees forward.
(b) 45 degrees backward.
(c) 45 degrees to the left.
(d) 45 degrees to the right.

In each position a referenced measurement (see 4.8.2) shall be made. Performance shall conform to 3.11.6.

4.8.7 Supply voltage and frequency (steady state). The transducer shall be operated at the configurations of normal, maximum and minimum steady state voltages 3.6.8. The ambient temperature shall also vary with the transducer operated for at least 15 minutes at each test temperature prior to the first reference measurement. The transducer shall be allowed at least 15 minutes to stabilize at each configuration after which a reference measurement is to be taken (see 4.8.2). Reference measurements shall be performed at ambient temperatures of 0° ± 2°C, 25° ± 2°C and 65° ± 2°C. Test temperatures shall be accomplished by varying temperature in steps of 10°C each (30 minutes per step) until the desired ambient temperature is reached. Performance shall conform to 3.11.7.

4.8.8 Common mode pressure test. For this test, the transducer's rated pressure specified (see 6.3) shall be applied simultaneously to both pressure ports. The pressure at the low pressure port shall then be decreased in pressure increments specified in 4.8.2 to the specified transducer range and then increased in similar increments to the transducer rated pressure. Performance shall conform to 3.11.9.

4.8.9 Frequency response. The sine wave differential pressure input signal used shall have a peak to peak magnitude of 10 percent of the input span of the transducer tested. The sine wave shall have the arithmetical midpoint of the transducer span as its center. Magnitude ratio and phase angle shall be determined over a frequency range from 0.01 cps to 20 cps in at least 10 equally spaced intervals plotted on a logarithmic scale. Performance shall conform to 3.11.10.

4.8.10 Supply voltage and frequency (transient). Tests specified in 4.8.10.1 and 4.8.10.2 shall be conducted with a differential pressure input signal equal to 80 ± 5 percent of the transducer span. Performance shall conform to the requirements of 3.11.11.

4.8.10.1 Transient voltage.

4.8.10.1.1 With the transducer operating at the upper limit of steady state voltage, a transient voltage of plus 20 percent of nominal voltage recovering to the steady state band in 2 seconds shall be superimposed.

4.8.10.1.2 With the transducers operating at the lower limit of steady state voltage, a transient voltage of minus 20 percent of nominal voltage recovering to the steady state band in 2 seconds shall be superimposed.

4.8.10.2 Transient frequency. A transient frequency increase of 2 cps shall be applied of which no more than one-half cycle is outside of the steady state band. Recovering to the steady state band in 2 seconds. This shall be repeated as above with a minus 2 cps transient frequency. Performance shall conform to 3.11.11.

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4.8.11 Temperature. The transducer shall be capable of normal operation (without alignment or adjustment) throughout the following temperature cycle; tolerances in operating characteristics shall be as specified herein:

(a) Hold room temperature at 0° ± 2°C. for at least 24 hours.
(b) Increase room temperature in steps of 10° each, at 30 minutes per step, until +65° ± 2°C. is reached, and hold at that temperature for at least 4 hours.
(c) Reduce from temperature in steps of 10° each, at 30 minutes per step, until +25°C. ±2°C. is reached, and hold at that temperature for at least 4 hours.

At each temperature plateau (0°C, 65°C and 25°C), a reference measurement (see 4.8.2) shall be made. Performance shall conform to 3.11.12. Temperature error shall be as specified in 3.11.8.

4.8.12 Humidity. Transducer shall be subjected to the following conditioning and test. Except for the periods of test indicated below, the equipment shall not be energized.

4.8.12.1 Conditioning. In order to establish a reference condition for the measurement of operating parameters and a valid basis for comparison of the effects of the conditioning to follow, the complete equipment shall be dried at a temperature not less than 40°C. nor more than 50°C. for at least 2 hours.

4.8.12.2 Reference measurements. Following the conditioning specified in 4.8.12.1, at reference measurement in accordance with 4.8.2 to indicate the satisfactory performance of the transducer shall be conducted at 25° ± 5°C. and 50 ± 5 percent relative humidity.

4.8.12.3 Temperature cycling. The transducer shall then be subjected to five 24-hour cycles of temperature variation consisting of approximately 16 hours at 65° ± 5°C and approximately 8 hours at 25° + 5°C. The relative humidity shall be maintained above a minimum of 95 percent during the steady state conditions. The transitions between temperatures shall be accomplished within the 8-hour period so that the time at the high temperature is approximately 16 hours. Each transition shall not exceed 1-1/2 hours if the transducer remains in the chamber, or 15 minutes if a two chamber method is employed. The relative humidity need not be controlled during the transition periods. Approximately 2 hours after stabilization during the high temperature and low temperature portions of the first or second cycle, a sampling of the atmosphere in the chamber shall be made to determine that the conditions of temperature and relative humidity specified above are uniform throughout the chamber.

4.8.12.4 Measurement during cycling. During the second cycle, the measurement required in 4.8.12.2 shall be performed at 65° ± 5°C. prior to the decrease of temperature to 25°C. with the transducer remaining in the chamber. The transducer shall be energized for as brief a period as required to complete the measurements. A warm-up period may be permitted where previous tests indicate a definite period is required for the transducer to attain thermal stability.

4.8.12.4.1 The transducer shall be capable of meeting the requirements specified herein without alignment or adjustments, other than the accessible controls employed for operation of the equipment. No repairs shall be permitted prior to measurement. If repairs are required, the test specified in 4.8.12.3 shall be repeated, after the necessary replacements have been made.

4.8.12.5 Measurements after temperature cycling. After the 5 complete cycles, the measurements required in 4.8.12.2 shall be performed at 25° ± 5°C with the transducer remaining in the chamber. The transducer shall meet the requirements of this specification with no readjustments.

4.8.12.5.1 Upon completion of the tests and after remaining inoperative for not less than 12 nor more than 24 hours at a temperature of +25° ± 5°C. and 50 ± 5 percent relative humidity, any additional tests or measurements considered necessary by the Government shall be made to determine complete compliance with the requirements specified herein.

4.8.12.6 Humidity cycle. The transducer shall then be operated continuously at full power in an ambient temperature of +50° ± 5°C. for a period of 8 hours. The relative humidity shall be increased from 50 ± 5 percent during the first 2 hours, maintained at 95 ± 5 percent for 4 hours, then reduced to 40 ± 5 percent during the final 2 hours. Measurements as specified herein shall be taken at 30 minutes intervals during this 8-hour period.

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4.8.12.7 The transducer shall be carefully examined in detail to detect evidence of physical degradation, such as corrosion of metal parts, distortion of plastic parts. When it is necessary to replace parts, to obtain satisfactory performance of the transducer the failed part or parts shall be analyzed to determine the cause of unsatisfactory operation. The results of the analysis shall be reported with the results of measurements of the transducer operating parameters. In each case, the unsatisfactory parts or material shall be replaced by adequate substitutes. Blistering for organic coating shall not be cause for rejection.

4.8.12.8 During the humidity cycle a pressure input equivalent to 80 ± 5 percent of the transducer span shall be maintained. At 30 minute intervals the pressure shall be decreased to the lower range limit, increased to the 80 ± 5 percent value and the transducer output shall be measured. Performance shall conform to 3.11.13.

4.8.13 Overpressure. Prior to the overpressure test, a reference measurement in accordance with 4.8.2 shall be made. The transducer shall successfully withstand differential pressure equal to 130 percent of the range for the period of 1 hour. At the end of this period, transducers shall be immediately subjected at a pressure equal to 1 psi or 10 percent of range, whichever is less, below atmospheric for an additional period of 1 hour. Within 10 minutes after release of this differential pressure a reference measurement (see 4.8.2) shall be made for comparison. Performance shall conform to 3.11.14.

4.8.14 Line pressure. The transducer shall successfully withstand the pressure rating specified (see 6.3) when applied to the high pressure port for a period of 10 minutes. The above shall be repeated with the pressure applied to the low pressure port of the transducer. After each test, a reference measurement in accordance with 4.8.2 shall be made. Performance shall conform to 3.11.15.

4.8.15 Enclosure. The enclosure shal be subjected to the splashproof test specified in MIL-STD-108. Performance shall conform to 3.11.24.

4.8.16 Salt spray. The salt spray test shall be conducted in accordance with method 101 of MIL-STD-202 using a 20 percent salt solution. Duration of the test shall be 100 hours. The internals of the signal conditioner and power supply shall not be subjected to the test. When any transducer component is physically attached to the detector, the integral component shall also be tested. Performance shall conform to 3.11.16.

4.8.17 Differential pressure whip. Prior to differential pressure whip test a reference measurement shall be made (see 4.8.2). The test shall be conducted on a suitable system capable of applying a sinusoidal pressure change of 20 to 80 percent of span for a total of 260,000 cycles. The rate of cycling shall be approximately 1 cps. The transducer shall be energized throughout the test. After completion of the differential pressure whip test a reference measurement shall be made for comparison (see 4.8.2). Performance shall conform to 3.11.17.

4.8.18 Dielectric strength. Dielectric strength between electrical input and output circuits and between these circuits and ground, shall be tested with a closely sinusoidal source of 60 cycles. Root mean square (rms) values of test voltage shall be as follows:

Circuit voltage of equipment tested	Rms value of dielectric test voltage
Less than 60	450
60 to 120	900

4.8.18.1 Equipment shall pass successfully, one high voltage test of applicable voltage specified in 4.8.18. In dielectric tests the voltage shall be raised gradually to the specified value and held at that value for the periods listed in 4.8.18.2. Performance shall conform to 3.11.18.

4.8.18.2 For the dielectric strength test conducted as part of qualification and periodic inspection, the test voltage shall be held at the specific value for 1 minute + 5 seconds minus zero seconds. For the dielectric strength test conducted as part of quality conformance inspection, the test voltage shall be held at the specified value for 5 ± 1 second minus 0 seconds.

4.8.19 Insulation resistance. The insulation resistance of the transducer shall be determined by applying 500 volts dc between electrical input and output circuits and between these circuits and ground at approximately room temperature 25° ± 2°C and at a relative humidity of 50 ± 5 percent (see 3.11.19).

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4.8.20 Vibration. The transducer shall be tested in accordance with type I (environmental vibration) of MIL-STD-167 except that the upper limit of the frequency range will be 100 cps. The amplitudes of vibration shall be in accordance with table VI. If no resonances are observed, the 2 hour endurance test shall be conducted at 100 cps. During the vibration test a liquid pressure equal to 80 ± 5 percent of the transducer span shall be applied to the transducer. Transducer output during the test shall be monitored. Performance shall conform to 3.11.20.

Table VI - Amplitudes of vibration.

Frequency range (cps)	Table amplitude Inch
5 to 20	0.030 ± 0.006
21 to 50	.020 ± 0.004
51 to 100	.010 ± 0.002

4.8.21 Shock. The shock tests shall be conducted in accordance with grade A, class 1, type C of MIL-S-901. During the test, a liquid differential pressure input equal to 80 ± 5 percent of the transducer span shall be applied to the transducer. Transducer output during the test shall be monitored. Before and after the shock test, reference measurements shall be made for comparison (see 4.8.2). Performance shall conform to 3.11.21. After completion of the shock test, the dielectric strength test specified in 4.8.18 shall be performed at 65 percent of the test voltage specified therein.

4.8.22 Burst pressure. The transducer shall be subjected to a liquid pressure equal to 200 percent of the pressure rating, with a maximum of 8000 psi., applied simultaneously to both sides of the transducer for a period of 10 minutes. The transducer shall conform to the requirements of 3.11.22. No performance tests shall be required after the burst pressure test. A reference measurement (see 4.8.2) shall be recorded for information purposes.

4.8.23 Output. A reference measurement shall be made in accordance with 4.8.2. Performance shall conform to the requirements of 3.11.23.

4.8.24 Temperature error. The temperature error shall be determined from the data obtained in 4.8.11. Performance shall conform to 3.11.8.

4.9 Inspection for preparation for delivery. The packaging, packing and marking shall be inspected for compliance with section 5 herein.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.5).

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts (see 5.3.1).

5.1.1 Differential pressure transducers.

5.1.1.1 Preservation and packaging. Preservation and packaging which may be the supplier's commercial practice, shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply soruce to the using activity and until early installation.

5.1.1.2 Packing. Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damge during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules or other carrier regulations as applicable to the mode of transportation and may conform to the supplier's commercial practice.

5.1.1.3 Marking. Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

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5.1.2 Onboard repair parts. Repair parts for onboard stowage and future use shall be packaged level A, packed level C and marked level A and C respectively in accordance with the applicable requirements of this specification. For submarines, packaging methods shall be modified in accordance with MIL-STD-758.

5.2 Domestic shipment and storage or overseas shipment. The requirements, and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 5.3.2 and 6.3).

(5.2.1 The following provides various levels for protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made.

5.2.1.1 Preservation and packaging.

5.2.1.1.1 Level A.

5.2.1.1.1.1 Cleaning. Unless otherwise specified herein, prior to packaging, transducers shall be cleaned in accordance with process C-1 of MIL-P-116.

5.2.1.1.1.2 Packaging.

5.2.1.1.1.2.1 Differential pressure transducers. Transducers shall be individually cushioned and packaged in accordance with method III of MIL-P-116 one to a box (see 5.2.1.1.1.2.2).

5.2.1.1.1.2.2 Unit containers. Unit containers shall conform to any of the following containers at the option of the contractor:

Specification	Classification
PPP-B-566	Variety 4, style 1
PPP-B-636	Type CF, weather resistant
PPP-B-665	Style A
PPP-B-676	Type I, class A

Container closure and sealing shall be in accordance with the applicable container specification or appendix thereto.

5.2.1.1.2 Level C. Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage. This level may conform to the supplier's commercial practice.

5.2.1.2 Packing. Packing shall be level A or B, as specified in the contract or order.

5.2.1.2.1 Level A. Transducers packaged as specified (see 5.2.1.1.1.2) shall be packed in weather-resistant fiberboard containers in accordance with PPP-B-636 or class 2, fiberboard-corrugated triple-wall containers in accordance with PPP-B-640, at the contractor's option. All center and edge seams and manufacturer's joints shall be sealed and waterproofed with pressure sensitive tape in accordance with the applicable fiberboard box specifications or appendix thereto. Shipping containers shall be closed and banded in accordance with the applicable box specification or appendix thereto. Banding (reinforced) shall be accomplished with use of pressure sensitive, reinforced tape. The gross weight of triple-wall boxes should not exceed 250 pounds. PPP-B-636 containers shall not exceed the weight limitations of the specifications. When specified, unit fiberboard containers conforming to weather resistant class of PPP-B-636 closed, sealed and banded as specified herein may be used as the shipping container and need not be over packed.

5.2.1.2.2 Level B. Transducers packaged as specified (see 5.2.1.1.1.2) shall be packed in containers conforming to PPP-B-636, domestic type or PPP-B-640, class 1. Shipping containers shall be closed in accordance with the applicable container specification or appendix thereto, except that fiberboard containers may be sealed with PPP-T-45 tape. Shipping containers shall not exceed the weight limitations of the applicable box specification. Unit fiberboard containers conforming to PPP-B-636 closed as specified herein may be used as the shipping container and need not be overpacked.

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5.2.1.3 Marking. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked for shipment in accordance with MIL-STD-129.

5.2.2 Repair parts (submarine, onboard and stock). Level A preservation and packaging methods shall be modified in accordance with MIL-STD-758 for items that can be stored in bins or drawers.)

5.3 Use of polystyrene (loose-fill) material.

5.3.1 For domestic shipment and early equipment installation and level C packaging and packing. Unless otherwise approved by the procuring activity (see 6.3), use of polystyrene (loose-fill) material for domestic shipment and early equipment installation and level C packaging and packing applications such as cushioning, filler and dunnage is prohibited. When approved, unit packages and containers (interior and exterior) shall be marked and labelled as follows:

"CAUTION

Contents cushioned etc. with polystyrene (loose-fill) material.
Not to be taken aboard ship.
Remove and discard loose-fill material before shipboard storage.
If required, recushion with cellulosic material bound fiber, fiberboard or transparent flexible cellular material."

5.3.2 For level A packaging and level A and B packing. Use of polystyrene (loose-fill) material is prohibited for level A packaging and level A and B packing applications such as cushioning, filler and dunnage.

6. NOTES

6.1 Intended use. Transducers are intended to provide control input and monitoring of differential pressures for shipboard engineering plants.

6.2 Attention of design engineers is invited to the following, which should be directed to NAVSEC for approval:

(a) Use of oil (see 3.6.12)
(b) Nonstandard-semiconductors and integrated circuits (see 3.7.2)
(c) Modular assemblies (see 3.7.4)
(d) Drawings and drawing revisions (see 3.8)
(e) Manuals (see 3.9)
(f) Periodic test reports (see 4.5.1)

6.3 Ordering data. Procurement documents specify the following:

(a) Title, number and date of this specification.
(b) Part classification required (see 1.2.1).
(c) Pressure rating (see 1.2.1.7).
(d) Transducer differential pressure range (see 3.5.3).
(e) Type of enclosure mounting (see 3.6.4).
(f) Type of differential pressure connections (see 3.6.7).
(g) Nominal power supply (see 3.6.8).
(h) Number of drawings (see 3.8.2).
(i) Quantity of manuals (see 3.9).
(j) Onboard repair parts required (see 3.10).
(k) Quantity of transducer required (see 4.5).
(l) Preservation, packaging, packing and marking required, if other than specified in 5.1 (see 5.2).
(m) When polystyrene (loose-fill) material is approved (see 5.3).
(n) Requirements for externally mounted components.

6.4 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in applicable Qualified Products List QPL 24304 whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Ship Engineering Center, Department of the Navy, Center Building, Prince George's Center, Hyattsville, Maryland 20782, and information pertaining to

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qualification of products may be obtained from that activity. Application for qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.4.1).

6.4.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

6.5 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in Section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.6 CHANGES FROM PREVIOUS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, ETC.) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT AS WRITTEN IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

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Naval Ship Engineering Center
Center Building
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Hyattsville, Maryland 20782

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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